QP CODE: 26900075

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours) / MGU-BBA (Honours) / MGU-BCA (Honours)
REGULAR EXAMINATION – OCTOBER 2026
FIFTH SEMESTER
Discipline Specific Elective (DSE) – MG5DSEBOTA01 - AGROBOTANY OF PLANTATION CROPS AND SPICES
(2024 ADMISSION ONWARDS)

ANSWER KEY

Duration: 2 Hours Maximum Marks: 70
Remember(k), Understand (u), Apply (A), Analyse(An), Evaluate (E), Create(C), Skills(S), Interest (I) and Appreciation (Ap)
Students should attempt atleast one question from each course outcome to enhance their overall outcome attainabilty

PART A
SHORT ANSWER
Answer any 10 questions. Each question carries 2 marks.(2 X 10= 20)

1. Name any two plantation crops that are export-oriented. [K]/ [CO1]
Tea, coffee. (Any other)

2. Identify the major plantation crop in India that is important for export. [K]/ [CO1]
Tea, coffee. (Any other)

3. Mention the role of spices in India's foreign exchange earnings. [K]/ [CO1]
Exports and contribute to agricultural trade (Any other).

4. Write one medicinal use of clove. [K]/ [CO2]
Temporary relief of toothache , analgesic , antiseptic properties,Digestive support,Pain and inflammation relief,Respiratory aid,Antimicrobial and immune defence (Any other)

5. Name one tool used for propagation observed during the field visit. [K]/ [CO1]
Budding/grafting knife, secateurs, pruning shears or a grafting tool, Hand Trowel, Dibble, Grafting tape, Knife (Any other)

6. Summarize the watering requirements of rubber seedlings. [U]/ [CO4]
Keep the nursery medium moist but not waterlogged, Good drainage to prevent root damage.

7. Explain the role of improved irrigation practices in supporting high-yield varieties. [U]/ [CO4]
Reduces moisture stress , improves nutrient uptake. Drip or scheduled irrigation can increase yield and water-use efficiency.

8. Explain what you learned about seed selection and its importance in crop productivity. [U]/ [CO4]
Select healthy, mature, viable seeds from disease-free plants,Good-quality seed improves germination, uniform crop establishment, plant vigour and yield, reducing the risk of seed-borne diseases.

9. Name any two micronutrients required for coffee. [K] /[CO1]
Zinc (Zn) , Boron (B) , Iron, Manganese, Copper.

10. Name two organic manures commonly applied in plantation crops like coconut and tea. [K] /[CO1]
Farm Yard Manure (FYM) ,Compost / Coir pith compost , Green leaf manure , Neem cake , Vermicompost.

11. Name the major insect pest responsible for boring into coffee berries. [K] /[CO3]
Coffee Berry Borer

12. Describe berry disease in coffee plantations. [U] /[CO3]
Caused by Colletotrichum kahawae. Symptoms- Dark brown-black sunken lesions and spot on green berries, pink spore masses, mummification and fall.

PART B
SHORT ESSAY
Answer any 8 question. Each question carries 5 mark (5X8=40)

13. Classify flower spices based on plant part used and provide suitable examples for each category. [U] /[CO3]
A) Flower Bud - Unopened Flower Bud- Clove (Syzygium aromaticum).
B) Stigma and Style - Saffron (Crocus sativus)

14. Describe the Importance of resistance to pests and diseases in coffee cultivar selection. [U] /[CO3]
Saves pesticide cost ,Protects against Leaf Rust, Coffee Berry Disease, Stem Borer. Reduces residues, suitable for organic/export, stable yield, reduse use of pesticides, increase farmers income, sustainable cultivation.

15. Explain the Botanical features of tea plants and their role in tea production. [U] /[CO2]
Camellia sinensis, Theaceae. Evergreen shrub maintained as 1-1.5 m bush by pruning, Deep tap root, Leaves - economic part - simple, elliptic, serrated, two leaves and a bud rich in caffeine , Flowers white, self-incompatible. Fruit capsule.

16. list the desirable characteristics of rubber cultivars [K] /[CO1]

High yield , early tappability, growth, disease resistance to Phytophthora, Oidium, wind resistance, drought tolerance, thick smooth bark with good renewal, timber value, good quality rubber. Clones: RRII 105 (70% area), RRII 414, RRII 430,RRIM 600

17. List the characteristics of good mother palm for coconut seed nut collection. [K] /[CO1]

regular nut yield, healthy and vigourous growth , produce large sized nut with good copra content, early bearing and regular flowering, free from bud rot, root wilt, stem bleeding, Ganoderma, rhinoceros beetle, Straight stout trunk, closely spaced leaf scars, spherical crown with 30-40 leaves, short fronds, adapted to local soil and climatic conditions.

18. Explain the process of callus induction and plant regeneration in rubber tissue culture. [U] /[CO4]

Methods-Somatic embryogenesis ,Micropropagation
Explants-Immature anthers ,Leaves ,Shoot tips
Importance-Clonal multiplication ,Disease-free stock
Explant selection, surface sterilization, culture medium(MS) preparation, inoculation, callus induction and multiplication, embryo induction, plant regeneration, hardening and field transfer.

19. Explain how fertilizers are applied at different stages of tea plant growth. [U] /[CO3]

Nursery stage – balanced fertilizers to Support seedling growth- FYM , phosphate , Urea monthly. Young tea – NPK
Mature tea 4 yrs onwards NPK
Pruned year – NPK , add required fertilizers for shoot development
Harvesting stage- nitrogen based fertilizer to support continuous leaf production.
Apply fertilizer when soil is moist , keep them away from stem and water is necessary.

20. State the Geographical and environmental conditions suitable for rubber plantation. [K] /[CO1]

Temperature -25–35°C
Rainfall - 2000–3000 mm/year
Relative humidity -75–95%
Altitude -Up to 500 m
Sunshine-5–7 hours/day
Climate – hot and humid
Soil – loamy soil

21. Describe the relationship between proper postharvest handling, quality control, and shelf life of plantation crops. [U] /[CO5]

Post Harvest Handling: Harvest at right maturity (coffee red cherry, tea two leaves and bud, coconut 12 months, rubber early morning), quick transport, proper processing (washing, withering, drying, smoking, mechanical bruishing, grading, sorting, drying, proper packing).

Quality Control: Drying to safe moisture ,prevents Aspergillus Ochratoxin, grading size/density, vacuum packing, Agmark certification, QC establishes measurable standard - moisture content, bean/leaf size, color, oil content, chemical purity, and absence of forgein matter or mycotoxins .

Shelf life:Harvest at correct maturity, quick pre – processing, proper drying,cleaning and grading, scientific storage, value addition , appropriate preservatiobn methods, inspect product regularly.

22. Differentiate between broad-spectrum and selective pesticides. [U] /[CO5]

Broad-spectrum - kills wide range of pests+beneficials, e.g Malathion, Mancozeb, Bordeaux. Kills pollinators, not IPM compatible, disturb natural ecosystem, biomagnification, cheap quick action.

Selective - kills only specific target, e.g. Bt for caterpillars , Neem, pheromones. Safe for beneficials like ladybird, bees, Integrated Pest Managment compatible , biodegradable , maintain ecological balance.

Part C

Essays

Answer any 1 question. Each question carries 10 marks.(10X1=10)

23. Analyze the nutrient management practices in coconut cultivation by comparing conventional (chemical-based) and organic (biofertilizer-based) approaches. [An] /[CO3]

Nutrient management in coconut aims to achieve high yield, maintain soil fertility, and ensure long-term sustainability. It can be broadly classified into conventional and organic methods.

1. Conventional Method (Chemical-Based)

Nutrient sources:

Chemical fertilizers - urea

Application method:

Adult palms (5 years and above) receive FYM per palm annually along with chemical fertilizers, Fertilizers are applied in circular basins around the palm in split doses, usually before and after the monsoon, applied through drip fertigation.

Advantages: Nutrients are quickly available to plants, Corrects nutrient deficiencies rapidly, Supports high yield and productivity, Requires less material because fertilizers are concentrated.

Disadvantages: Excessive use cause soil acidification ,reduce organic matter, negatively affect beneficial soil microbes and earthworms, nutrient leaching and runoff may pollute water bodies.

2. **Organic Method (Manure- and Biofertilizer-Based)**

Nutrient sources:

- FYM, compost, vermicompost, neem cake, and coir pith compost.
- Green manures .
- Biofertilizers.

Application method:

Applied in basins or trenches around the palm, Coconut leaves are used as mulch to conserve soil moisture, Biofertilizers are mixed with compost and applied near the active root zone.

Advantages: Improves soil structure, aeration, and water-holding capacity, Increases beneficial soil microorganisms, Helps in nitrogen fixation and phosphorus availability, Increases soil organic matter and carbon storage, Reduces dependence on synthetic fertilizers and supports sustainable cultivation.

Disadvantages: Nutrients are released slowly, Requires large quantities of manure and compost, Needs more labour for composting, mulching, and green manuring.

24. Explain significance of spice-based nutraceuticals in preventive healthcare with examples from Cardamom, Black pepper, Cloves, Turmeric. [U] /[CO5]

Nutraceutical = Nutrition + Pharmaceutical. Bioactive compounds from spices providing health benefits beyond nutrition, preventing chronic diseases. Low side effects.

1. **Cardamom** (Elettaria cardamomum, Zingiberaceae) - Queen of Spices:

Bioactive: 1,8-Cineole, Cardamonin.

Uses: Digestive - carminative, Irritable Bowel Syndrome, stimulates enzymes, Antioxidant - prevents cardiovascular disease. Anti-diabetic - improves insulin sensitivity. Oral health – Cavities,gum diseases, bad breath. Products: Cardamom oil capsules, tea.

2. **Black Pepper** (Piper nigrum, Piperaceae) - King of Spices:

Bioactive: Piperine

Significance: Bioenhancer, Neuroprotection, Gastro instestinal support, Anti-obesity - inhibits adipogenesis, thermogenesis. Anti-depressant - increases serotonin. Anti-inflammatory. Products: BioPerine capsules, weight loss supplements.

3. **Cloves** (Syzygium aromaticum, Myrtaceae) - Flower bud:

Bioactive: Eugenol

Significance: Dental - analgesic, antiseptic, root canal filling, toothache gel. Antioxidant value, prevents aging. Anti-diabetic - improves insulin, Anti-cancer. Products: Clove oil 85% eugenol, mouthwash, immunity tea.

4. **Turmeric** (Curcuma longa, Zingiberaceae) - Golden Spice:

Bioactive: Curcuminoids - Curcumin , Turmerone.

Significance: Anti-inflammatory , Anti-cancer. Cardio-protective - lowers glucose, Neuroprotective , Wound healing, immunomodulator. Gastrointestinal support. Products: Curcumin capsules, Golden milk latte, Curcumin-Piperine Complex, cosmetics.